



ANDERSON JUNIOR COLLEGE
2009 PRELIMINARY EXAMINATION
HIGHER 2

CHEMISTRY

9746/03

Paper 3 Free Response

14 September 2009

2 hours

Candidates answer on writing paper.

Additional Materials : Data Booklet
Graph Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PDG on all the work you hand in.
Write in dark blue or black pen on both sides of the paper. **Start each question on a fresh page.**
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each questions or part question.

At the end of the examination, detach this cover page and attach to the top of your answer scripts securely together.

In the table, circle the question number you've attempted.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	80

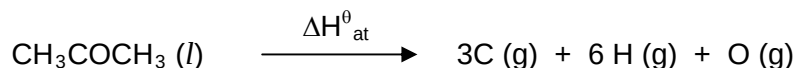
- 1 (a) The standard enthalpy change of combustion of liquid propanone, CH_3COCH_3 , is $-1790 \text{ kJ mol}^{-1}$.

In an experiment, 2.9 g of propanone was burned under a container of water, using a spirit lamp, to heat the water from 15°C to its boiling point. The process was found to be 70% efficient.

- (i) Write an equation, with state symbols, which represents the standard enthalpy change of combustion of propanone.
- (ii) Calculate the mass of water that could be brought to the boiling point by burning this amount of propanone.
[Given specific heat capacity of water is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$]
- (iii) Suggest a reason why the process is only 70% efficient.
- (iv) Explain why the above combustion has a negative entropy change.
- (v) Is the reaction feasible at high temperature? Explain your answer.

[8]

- (b) The standard enthalpy change of atomisation of propanone, $\Delta H^\theta_{\text{at}}$, is represented by the equation:



- (i) By using the following data, draw an appropriate energy cycle and calculate the enthalpy change of atomisation of propanone.

enthalpy change of formation of propanone	$= -248 \text{ kJ mol}^{-1}$
enthalpy change of atomisation of carbon	$= +715 \text{ kJ mol}^{-1}$
enthalpy change of atomisation of hydrogen	$= +218 \text{ kJ mol}^{-1}$
enthalpy change of atomisation of oxygen	$= +249 \text{ kJ mol}^{-1}$
- (ii) By using appropriate bond energy data from the *Data Booklet*, calculate another value for the enthalpy change of atomisation of propanone.
- (iii) Hence calculate the enthalpy change of vaporisation of propanone.

[6]

(c) A student was given 4 unlabelled bottles and each bottle contains one of the following organic substances.

- CH_3COCH_3
- $\text{CH}_3\text{CH}_2\text{COO}^- \text{NH}_4^+$
- $\text{C}_4\text{H}_9\text{CHO}$
- $\text{CH}_3\text{CH}=\text{CHCH}_2(\text{OH})$

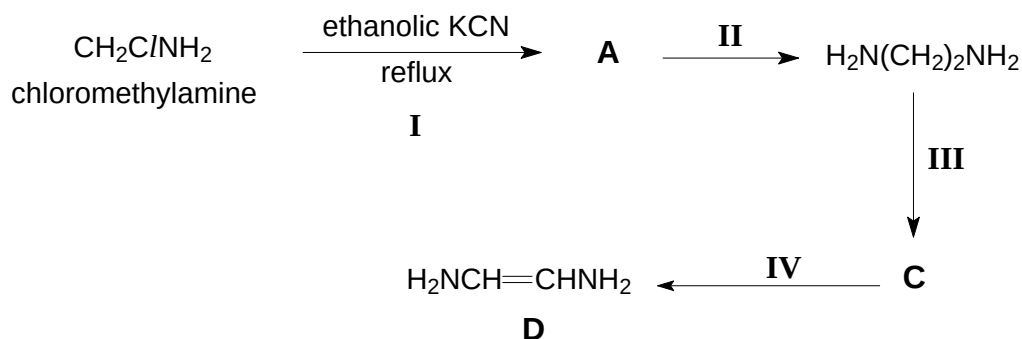
Outline a sequence of simple chemical tests which the student could use to identify **each** of the above organic substances. For each test, state the reagents and conditions used and describe the expected observations.

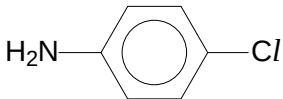
You are **not** allowed to identify the substances by elimination.
You may use a flow chart to present your answer.

[6]

[Total: 20]

- 2 (a) This question is about the reaction of chloromethylamine, $\text{CH}_2\text{Cl}/\text{NH}_2$.



- (i) Describe the reaction mechanism in step **I**. Hence deduce the identity of compound **A**.
In your answer, show any relevant charges, dipoles or lone pairs of electrons you consider to be important.
- (ii) State the reagents and conditions for steps **II**, **III** and **IV**.
- (iii) Suggest the identity of compound **C**. [6]
- (b) Describe a chemical test (one in each case) you could use to distinguish chloromethylamine from compounds **E** and **F** below. Suggest a different method for each test, giving reagents, conditions and observations.
- E** CH_2BrNH_2
- F** 
- [5]
- (c) Suggest why compound **F** does not undergo the reaction in step **I**. [1]

- (d) One structural isomer of compound **D** is azomethane, $\text{CH}_3\text{N}=\text{NCH}_3$.

When heated, azomethane decomposes to ethane and nitrogen. The equation for the reaction is shown below.



This homogeneous reaction was followed experimentally by measuring the change in total volume of gas.

Time, t / s	0	210	420	860	1350	1800
Total Volume of gas, V / cm^3	600	606	610	615	618	619

- (i) Show that at any time t , the volume of N_2 gas is $(V - 600) \text{ cm}^3$.
- (ii) By plotting a graph of the volume of N_2 against time, determine the order of this reaction. Show your working clearly on the graph.
- (iii) Deduce the value of the rate constant.
- (e) With the aid of a suitable diagram, explain how the rate of reaction will be affected when this reaction is done at a higher temperature.

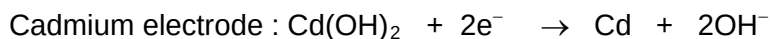
[5]

[3]

[Total: 20]

- 3 The nickel–cadmium (NiCd) battery and the dry cell battery are commonly used in portable devices such as handphones and torchlights.

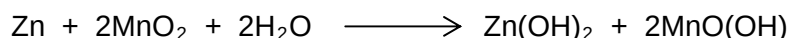
In the NiCd battery, the nickel and cadmium electrodes, both coated with their respective hydroxides, are immersed in aqueous potassium hydroxide. During the charging process, the following reaction occurs.



- (a) (i) Write a balanced equation for the reaction that occurs during charging.
- (ii) When a current of 3.0 A is used to charge a NiCd battery, the mass of Cd(OH)_2 decreases by 3.65 g. Calculate the time taken to charge the battery.
- (iii) Sealed NiCd batteries may develop high internal gas pressure when subjected to excessive charging. Using data from the *Data Booklet*, write balanced half–equations to identify the gas that is formed at each electrode.

[6]

- (b) In contrast to the NiCd battery, a dry cell battery is a non–rechargeable battery, with zinc and carbon electrodes. Solid MnO_2 is coated on the carbon electrode. The overall equation for the discharging reaction is given below.



- (i) Explain, in terms of changes in oxidation number, which electrode acts as the cathode.
- (ii) The e.m.f. of this cell is 1.60 V. Calculate the $E^\ominus$ value of the reduction reaction.
- (iii) State what happens to the e.m.f. of this dry cell battery when the size of the zinc electrode is decreased.
- (iv) State one advantage of using this dry cell battery rather than a NiCd battery.

[4]

- (c) Explain why manganese forms compounds that are often coloured whereas compounds of zinc are usually white.

[4]

- (d) MnO_2 is the most common starting material for the production of compounds of manganese of other oxidation states.

Oxidation of MnO_2 in molten KOH with oxygen from the air produces a green compound, **G**. **G** contains potassium, manganese and oxygen only.

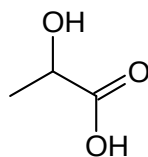
On treatment with dilute acid, a sample of compound **G** reforms 0.174 g of MnO_2 and a purple solution of compound **H**. A solution of compound **H** is just decolourised by the addition of 40.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ iron(II) sulfate.

- (i) Suggest the identity of the purple solution, **H**.
- (ii) Name the type of reaction that occurs when dilute acid is added to compound **G**.
- (iii) Calculate the number of moles of MnO_2 and compound **H**.
- (iv) Using your answer in (d)(iii), calculate the oxidation number of manganese in compound **G** and hence deduce the formula of **G**.

[6]

[Total: 20]

- 4 (a) Muscle fatigue and cramps commonly experienced after a strenuous exercise are due to the build-up of lactic acid when oxygen-starved skeletal muscle starts to respire anaerobically. The structure of lactic acid, a monoprotic acid, is shown below.



The normal pH of the muscle cell is 7.10 but if the build-up of lactic acid continues, the pH is reduced to around 6.50.

Given that the pK_a of lactic acid is 3.86, calculate the molar concentration of lactic acid in the muscle cell at pH 6.50.

[2]

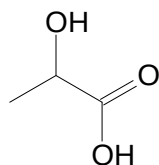
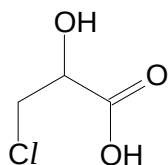
- (b) A buffer solution can be made by adding 22.4 g of solid sodium lactate to 1.0 dm^3 of 0.10 mol dm^{-3} lactic acid.

(i) Calculate the pH of this solution.

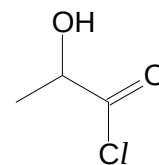
(ii) Calculate the pH of the buffer solution after the addition of 10.0 cm^3 of 0.10 mol dm^{-3} HCl.

[5]

- (c) Arrange the following compounds in order of **increasing** pH in aqueous solution. Explain your answer.



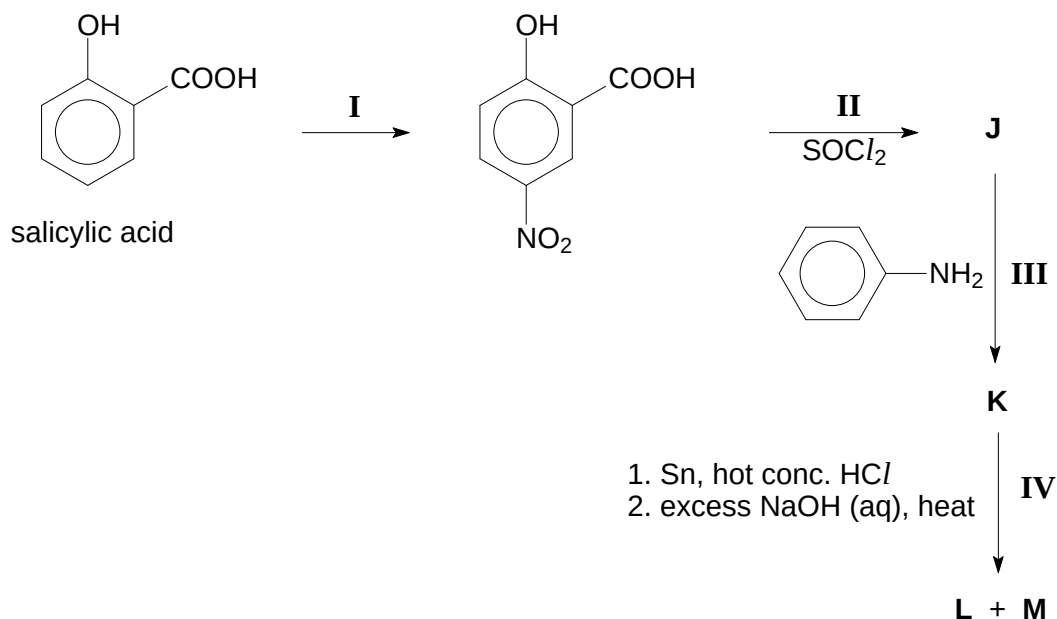
lactic acid



[3]

- (d) Salicylic acid, which is commonly found in facial cleansers, is helpful in treating acne. Like lactic acid, salicylic acid contains both a hydroxyl and a carboxylic acid group.

A reaction scheme of salicylic acid is shown below.



- State the reagents and conditions for step **I**.
- Suggest the type of reaction in step **I**.
- Give the structural formulae of compounds **J**, **K**, **L** and **M**.

[6]

- (e) Solid NaOH is slowly added to a solution containing 0.010 mol dm⁻³ Mg²⁺ and 0.010 mol dm⁻³ Fe²⁺.

$$K_{sp} \text{ of } \text{Mg}(\text{OH})_2 = 1.2 \times 10^{-11} \text{ mol}^3 \text{ dm}^{-9}$$

$$K_{sp} \text{ of } \text{Fe}(\text{OH})_2 = 1.6 \times 10^{-14} \text{ mol}^3 \text{ dm}^{-9}$$

- Calculate the pH at which the first trace of Fe(OH)₂ is formed.
- Calculate the pH at which the maximum amount of Fe²⁺ can be extracted from the solution. Hence, calculate the percentage of Fe²⁺ remaining in the solution at this point.

[4]

[Total: 20]

- 5 Inorganic and organic halogen compounds are commercially important industrial chemicals. Millions of tonnes of chlorine and hydrogen chloride are produced annually. Most of the chlorine is used in the manufacture of organic compounds such as PVC plastics, as well as in producing bleaching agents such as sodium chlorate(I) for paper and textile industries.

(a) (i) Describe with the aid of equations how chlorate(I), ClO^- and chlorate(V), ClO_3^- , can be obtained from chlorine in the laboratory.

(ii) Bromate(V), BrO_3^- , can be similarly obtained from bromine.

When hydrogen sulphide gas is bubbled into an acidified solution of potassium bromate(V), KBrO_3 , a yellow precipitate and an orange-red solution is obtained. On adding aqueous potassium iodide to the orange-red solution, the solution turned dark brown and some black solids are formed.

Explain the observations and identify the products of the reactions.

[5]

(b) Give reasons to explain why the boiling point of hydrogen chloride is lower than that of hydrogen fluoride and hydrogen iodide.

[2]

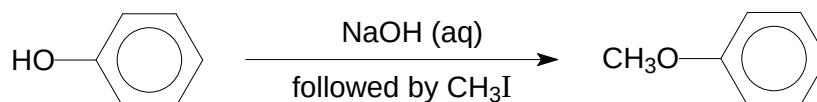
(c) Bromine trifluoride, BrF_3 , is a strong fluorinating agent. It reacts with antimony pentafluoride, SbF_5 , to form a salt; $(\text{BrF}_2^+)(\text{SbF}_6^-)$.

Describe the changes in the shapes of the molecules to their corresponding ions.

[4]

(d) Organic halogen compounds are widely used in synthetic reactions.

Iodoalkanes are often used in the synthesis of ethers from phenols in the Williamson ether synthesis, an example of which is shown below.

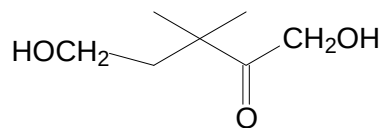


(i) Suggest why aqueous NaOH is required in this reaction.

(ii) The rate of the synthesis decreases when CH_3Cl is used in place of CH_3I . Suggest an explanation for this difference in the rate of reaction.

[3]

- (e) When aqueous bromine is added to an organic dyestuff **P**, $C_{10}H_{14}O$, bromine is decolourised and compound **Q**, $C_{10}H_{14}OBr_4$, is formed. **P** is insoluble in aqueous NaOH. Both **P** and **Q** give silver precipitates when reacted with Tollen's reagent. When **P** is treated with hot acidified potassium manganate(VII), two organic products, **R**, $C_3H_4O_3$, and **S**, $C_7H_{10}O_5$, are formed. **R** gives a pale yellow precipitate when warmed with alkaline aqueous iodine. Treatment of **S** with $LiAlH_4$ produces a compound with the structure below.



Suggest the structures of **P**, **Q**, **R** and **S**, giving your reasoning.

[6]

[Total: 20]